

DON PHU HUU TRAN



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RESEARCH INTEREST

- Cancer Biology
- Machine Learning
- Bioinformatics
- Quantum Foundation
- Quantum Information
- Quantum Darwinism

EDUCATIONS

MSc in Theoretical Physics – Mathematical Physics

2026 - present

Faculty of Physics and Engineering Physics

University of Science, Viet Nam National University

Ho Chi Minh city, Viet Nam

BSc in Molecular Biology

2022 - 2026

Faculty of Biology and Biotechnology

University of Science, Viet Nam National University

Ho Chi Minh city, Viet Nam

RESEARCH EXPERIENCES

Research student

2026 - present

Quantum Lab - HCMUS

University of Science, Viet Nam National University - HCM

Ho Chi Minh city, Viet Nam

Supervisor: Dr. Trần Nguyên Lâm – Dr. Lê Bin Ho

- Modeling system-environment interaction in Quantum Darwinism picture using qubits simulation.
- Simulating theoretical physics experiment using IBM Quantum hardware.

Research assistant

2025 - present

Research Center of Genetics and Reproductive Health (CRG

University of Health Science, Viet Nam National University - HCM

Ho Chi Minh city, Viet Nam

Supervisor: Assoc. Prof. Dr. Nguyễn Minh Nam

- Applied feature selection methods (LASSO, Boruta, Elastic Net, RFECV) to identify promising biomarker for multi-cancer detection
- Developed machine learning and deep learning models (Random Forest, XGBoost, Neural Networks, ...) for multi-cancer classification using multi-omics datasets
- Achieved strong predictive performance (**AUC and accuracy > 0.90**) across validation datasets
- Led experimental design and manuscript preparation. Mentored undergraduate students in data analysis and machine learning workflows

PUBLICATIONS

- **Don P.H Tran**, et al. (2026). An integrative machine learning framework identifies a circulating five-miRNA signature for glioma detection. Submitted (2026)
- Van Nguyen, **Don P.H Tran**, et al. (2026). Circulating hsa-miR-23b-3p as a Novel Biomarker for Multi-Cancer Classification. Submitted to 11th International Conference in Vietnam on the Development of Biomedical Engineering.
- **Don P.H Tran**, et al. (2026). Integrative bioinformatic analysis identifies hsa-miR-92a-2-5p as a candidate circulating biomarker for glioma detection and prognosis. Submitted to 11th International Conference in Vietnam on the Development of Biomedical Engineering.
- **Don P.H Tran**, et al. (2026). A microRNA-based machine learning framework for pancreatic cancer detection and RNA-target drug discovery. Submitted to Frontiers in Pharmacology

SCIENCE ACTIVITIES

- Oral presentation – 1st Youth Science Conference 2026, University of Health Science, Vietnam National University, Ho Chi Minh city, Viet Nam (July. 2026)
- Oral presentation – 3rd Youth Science Conference 2026, University of Science, Vietnam National University, Ho Chi Minh city, Viet Nam (March. 2026)
- Oral presentation – The 3rd Rencontres de Quy Nhon International Conference on Molecular Diagnostics in Microbiology and Diseases 2025, Quy Nhon, Viet Nam (December. 2025)
- Poster presentation – Rencontres de Quy Nhon VIII: International Biology Conference 2025, Quy Nhon, Viet Nam (September. 2025)
- Poster presentation – 1st International Conference on Health Science, University of Health Science, Vietnam National University, Ho Chi Minh city, (July. 2025)
- Poster presentation – 20th Pacific Basin Consortium for Environment and Health International Conference, (May. 2025)

AWARDS

- Best oral presentation – 1st Youth Science Conference 2026, University of Health Science, Vietnam National University, Ho Chi Minh city, Vietnam **2026**
- Best oral presentation – 3rd Youth Science Conference 2026, University of Science, Vietnam National University, Ho Chi Minh city, Vietnam **2026**
- First prize – Poster Presentation, 20th pacific basin consortium for environment and health, Ho Chi Minh city, Vietnam **2025**

MISCELLANEOUS

Language

- Vietnamese (native)
- English (fluently)
- Germany (beginner)

Programming language

- Python (scikit-learn, pandas, numpy, qiskit, ...)
- R (limma, DESeq2)